MATH 1 MIDTERM STUDY

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Answer 1:

$$\lim_{x\to\infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}} \Rightarrow \lim_{x\to\infty} \frac{\sin(\frac{1}{\infty})}{\frac{1}{\infty}} = \frac{\sin 0}{0} = \frac{0}{0}$$

$$\lim_{x\to\infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}} \Rightarrow y = \frac{1}{x} \Rightarrow \lim_{x\to\infty} \frac{\sin a}{a} = 1 //$$

Answer 2:

$$\lim_{x\to 1^+} \frac{|x-1| + [[x-3]]}{x+1} \Rightarrow 1^+ \Rightarrow |x-1| > 0 \Rightarrow (x-1) \Rightarrow 1-1=0$$

$$[[x-3]] \Rightarrow -2$$

$$\Rightarrow \frac{-2}{2} = -1 ///$$

Answer 3:

$$\lim_{x\to-\infty} \frac{\sqrt{5x^2+3x-2}}{3x+5} \Rightarrow \lim_{x\to-\infty} \frac{\sqrt{x^2(5+\cancel{\frac{3}{x}}^{0}-\cancel{\frac{2}{x^2}}^{0})}}{x\,(3+\cancel{\frac{5}{x}}^{0})} = \lim_{x\to-\infty} \frac{-\sqrt{5}\cdot\cancel{x}}{3\cancel{x}}$$

$$\Rightarrow \frac{-\sqrt{5}}{3} ///$$

Answer 4:

$$\lim_{x\to 0^+} \frac{1-\cos x}{x} \Rightarrow \lim_{x\to 0^+} \frac{1-\cos 0}{0} = \frac{0}{0}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{1-\cos x}{x} \cdot \frac{1+\cos x}{1+\cos x}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{1-\cos^2 x}{x.(1+\cos x)} = \lim_{x\to 0^+} \frac{\sin^2 x}{x.(1+\cos x)}$$

$$\Rightarrow \lim_{x\to 0^+} \frac{\sin x}{\underbrace{1+\cos x}_{2}} \cdot \frac{\sin x}{x} = \lim_{x\to 0^+} \frac{\sin x}{\underbrace{2x}_{\frac{1}{2}}} \cdot \underbrace{\sin x}_{0} = 0 ///$$

Answer 5:

$$\lim_{x\to-\infty}\frac{3x-\sqrt{5}}{\sqrt{9x^2+3x+1}} \Rightarrow \lim_{x\to-\infty}\frac{x\left(3-\cancel{\frac{\sqrt{5}}{x}}^{\,0}\right)}{\sqrt{x^2\left(9+\cancel{\frac{3}{x}}^{\,0}+\cancel{\frac{1}{x^2}}^{\,0}\right)}}$$

$$\Rightarrow \lim_{x\to-\infty}\frac{x\ 3}{x\ -\sqrt{9}}=\lim_{x\to-\infty}\left(-\frac{3}{3}\right)=-1$$

Answer 6:

$$\lim_{x\to-\infty}\frac{\sqrt{x^2+x+1}}{2x+5}=\lim_{x\to-\infty}\frac{\sqrt{x^2\left(1+\frac{1}{x}+\frac{1}{x^2}\right)}}{x\left(2+\frac{1}{x}\right)}$$

$$=\lim_{x\to-\infty}\frac{\cancel{x}\,(-1)}{\cancel{x}\ 2}=\frac{-1}{2}$$

Answer 7:

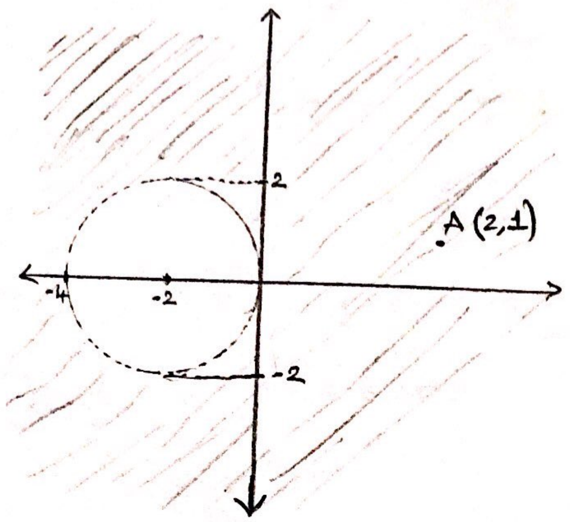
$$\lim_{x\to1^-}\frac{|x-1|+[\![x]\!]+2x}{\sqrt{3x}} \Rightarrow |x-1|<0 \Rightarrow -x+1$$

$$[\![x]\!]=[\![(0{,}999),(0{,}888)\ldots]\!]=0$$

$$\lim_{x\to1^-}\frac{(1-x)+0+2}{\sqrt{3}}=\frac{2}{\sqrt{3}}$$

2. sketch the graphs of the following relations on right coordinate system.

(a) $R=\{(x,y)\ (x+2)^2+y^2>4\}$

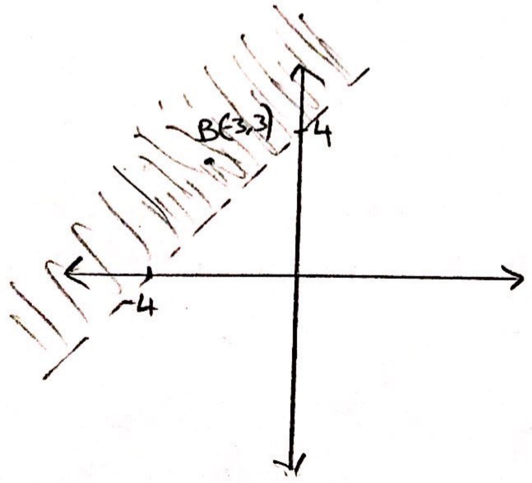


Theorem: $(x-a)^2+(y-b)^2=r^2$ is circle equation

$[x-(-2)]^2+(y-0)^2>2^2$

radius $=2$ center $=(a,b)=(-2,0)$

(b.) $S=\{(x,y)\mid y > x+4\}$

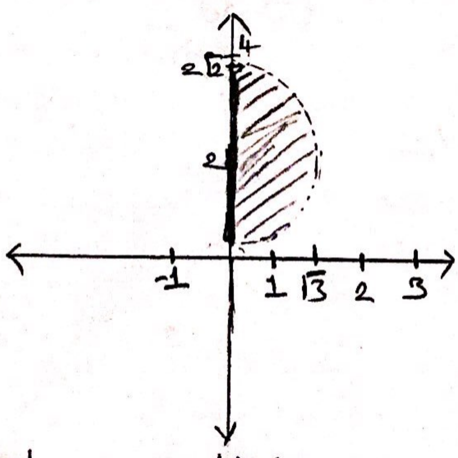


$y > x+4$

$y=0 \rightarrow x=-4$

$x=0 \rightarrow y=4$ $B(-3,3)$ satisfies the inequality

(c) $T=\{(x,y)\mid x^2+(y-2)^2<3,\ x\geq 0\}$



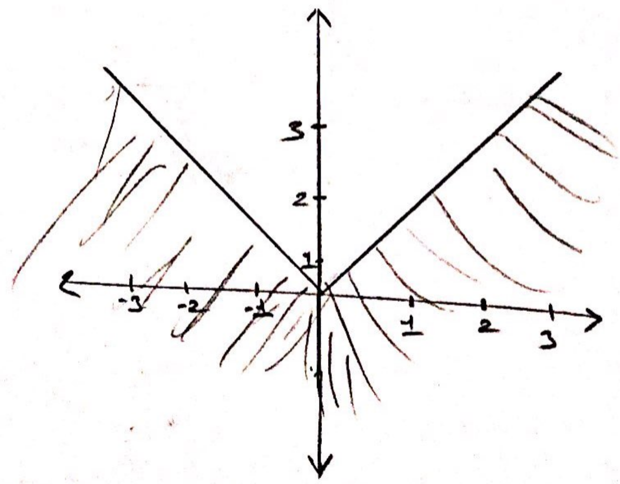
circle equation

$(x-0)^2+(y-2)^2<(\sqrt{3})^2$

radius $=\sqrt{3}$ center $=(0,2)$

But $x\geq 0$

(d) $U=\{(x,y)\mid y \leq |x|\}$



I-) $x\geq 0 \rightarrow |x|=x \rightarrow y\leq x$

x	0	1	2
y	0	1	2

II-) $x<0 \rightarrow |x|=-x \rightarrow y\leq -x$

x	-1	-2	-3
y	1	2	3

3. Find the solution set of the following inequality analyzing it in two cases $(2x+3 \neq 0)$

Case 1: $2x+3<0$

$x<\frac{-3}{2}$

$3x-2 \leq 2+3$

$x \leq 5$

$(-\infty, \frac{-3}{2})$

Case 2: $2x+3>0$

$2x+3>0$

$x>\frac{-3}{2}$

$3x-2>2x+3$

$x>5$

$(5,\infty)$

$R-[\frac{-3}{2},5]$